

MÖSSBAUER STUDIES ON THE GAMMA RADIOLYSIS OF CESIUM TRIS(OXALATO)FERRATE(III) DIHYDRATE

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The final products of the gamma radiolysis of cesium tris(oxalato)ferrate(III) dihydrate have been identified by Mössbauer spectroscopy and thermogravimetric analysis as a mixture of $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$ and $\text{Cs}_2\text{Fe}(\text{ox})_2$ in equivalent proportions. The tetrahydrated compound can also be made by wet synthesis whereas the anhydrous salt can be obtained from thermal decompositions of either the tetrahydrated ferrous salt at 160°C or of the ferric complex at 285°C . The radiolytic decomposition proceeds as a first-order process due to the original compound depletion and to the radiolytic stability of the radiolytic products.

1. Introduction

The gamma radiolysis of crystalline alkali tris(oxalato)ferrate(III) has been extensively studied using Mössbauer spectroscopy [1-7]. Though it was shown in all these studies that gamma irradiation always produces a reduction of iron(III) to iron(II), whatever the alkaline cation be, different interpretations of the Mössbauer spectra have been proposed for the identification of the radiolytic products.

If, for the sodium/1,6/ and the lithium/6,7/ hydrated complexes, it has been well established that the main radiolysis product is $\text{Feox} \cdot 2\text{H}_2\text{O}$, the Mössbauer technique revealed that this compound was clearly not produced in the case of the potassium salt for which the formation of several intermediates have been proposed, such as $\text{K}_6\text{Fe}_2(\text{ox})_5 \cdot 2,3,6/$, $\text{K}_2\text{Fe}(\text{ox})_2 \cdot 2\text{H}_2\text{O} \cdot 4/$ or a mixture of $\text{K}_4\text{Fe}(\text{ox})_3$ and $\text{K}_2\text{Fe}(\text{ox})_2 \cdot 8/$.

On the other hand and surprisingly, during the course of gamma radiolysis of the cesium complex $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$, B.S. RANDHAWA et al. [6] reported the formation of $\text{Feox} \cdot 2\text{H}_2\text{O}$, a result which seems beside the trend one would expect in this alkali complexes series.

For these reasons, it seemed interesting to us to re-investigate, more deeply, the effect of outer monovalent cations as well as of crystallization water on the radiolytic products of $\text{M}_3\text{Fe}(\text{ox})_3 \cdot x\text{H}_2\text{O}$ complexes ($\text{M} = \text{Cs}, \text{Rb}, \text{K}, \text{Na}, \text{Li}$). In the present work, the gamma radiolysis of cesium tris(oxalato)ferrate(III) dihydrate has been studied using Mössbauer spectroscopy and other chemical analysis methods in order to elucidate the exact nature of the radiolytic products as well as the mechanism of the radiolysis.

2. Experimental

The hydrated complex $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$ was synthesized in the conventional way and its composition was verified by thermogravimetric analysis (TGA) [9]. Samples for radiolysis experiments were pulverized until the particle size was 300-500 mesh.

Two model compounds, $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$ and $\text{Cs}_2\text{Fe}(\text{ox})_2$, were also prepared for comparison with the radiolytic products. $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$ was prepared by dissolving under nitrogen, solid $\text{Feox} \cdot 2\text{H}_2\text{O}$ in a boiling and concentrated (3.5M) cesium oxalate solution; after cooling at room temperature, the orange-yellow precipitate was filtered off and dried with alcohol. Anhydrous $\text{Cs}_2\text{Fe}(\text{ox})_2$ was obtained by dehydration of $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$ at 160°C and in oxygen-free nitrogen, in agreement with

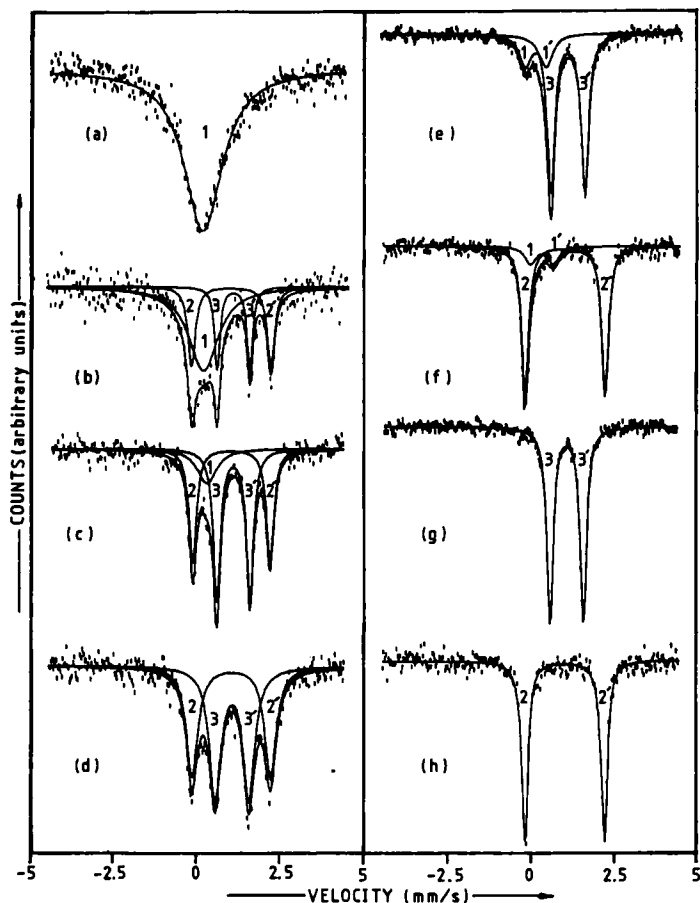


Fig.2. Mössbauer spectra at 298K of: a) unirradiated and b) c) d) irradiated $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$ up to a dose of 54 Mrad, 148 Mrad, 203 Mrad, e) sample c exposed to water vapour, f) sample c heated at 160°C , g) $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$, h) $\text{Cs}_2\text{Fe}(\text{ox})_2$.

at 160°C under nitrogen, the irradiated sample from fig.2c exhibits only one ferrous doublet (22') (Fig.2f) ($QS=2.35 \text{ mm/s}$; $IS=1.13 \text{ mm/s}$) corresponding to $\text{Cs}_2\text{Fe}(\text{ox})_2$, in agreement with the dehydration temperature of the tetrahydrated model compound. On the other hand, if this sample is exposed to atmospheric moisture, the tetrahydrated component (33') increases at the expense of the anhydrous one which disappears completely after a few days of exposition (Fig.2e), because of its hygroscopic character. It must also be mentioned that during these two treatments, the single line (1) has transformed into a ferric doublet (11') ($QS=0.64 \text{ mm/s}$, $IS=0.26 \text{ mm/s}$) with some increasing of intensity when exposed to moist air (Fig.2e); this is presumably due to a restructuring of the ferric complex and a partial oxidation of the ferrous species.

New evidences for this mechanism have been found recently [12] in the gamma radiolysis study of the anhydrous and the tetrahydrated cesium tris(oxalato) ferrate(III) for which the ferrous radiolytic products were found to be $\text{Cs}_2\text{Fe}(\text{ox})_2$ and $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$, respectively. It is therefore apparent that the crystallisation water content of the initial ferric complex determines the relative proportions of the two ferrous compounds in the irradiated residue, provided the samples are not exposed to moisture.

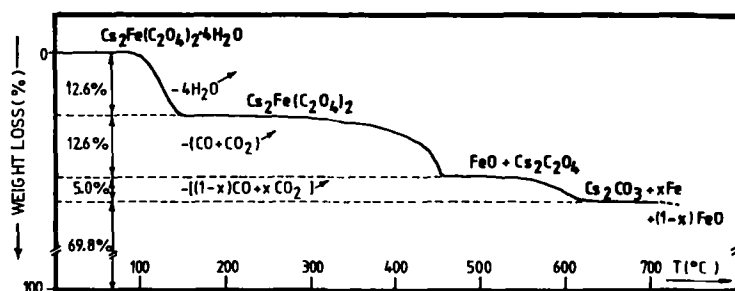


Fig.1. Thermogravimetric analysis of $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 4\text{H}_2\text{O}$ in nitrogen.

the relative weight loss observed (12.6%) on the TGA curve (Fig.1.) and the calculated value for the loss of $4\text{H}_2\text{O}$ (12.64%). As being readily oxidizable and hygroscopic, this anhydrous red salt was stored in dry nitrogen atmosphere. It should be noted that this compound is also formed by thermal decomposition of $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$ at $285^\circ\text{C}/9/$.

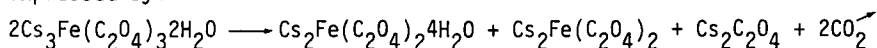
The powdered samples were irradiated at 298K under high purity nitrogen in sealed glass tubes. The irradiations have been performed using a 12000 Ci ^{60}Co source and the dose rate (0.56 Mrad/h) was calibrated by Fricke dosimetry. After irradiation, the samples were stored in a dry nitrogen atmosphere in order to prevent hydration or oxidation of the products before analysis and to allow the release of CO_2 still scavenged in the lattice. Potentiometric titrations of Fe^{3+} , Fe^{2+} and ox^{--} with Ce^{4+} and Cr^{++} solutions were then performed on several samples irradiated at increasing doses in order to follow the evolution of the radiolytic yield as a function of the absorbed dose.

Mössbauer measurements were made as previously described/11/. The spectra were recorded at 298K and 80K; all isomer shifts are quoted relative to natural iron.

3. Results and discussion

The spectrum of unirradiated $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$ (fig.2a) consists of a single broad line (1) due to relaxation effects/10/. The isomer shift value ($\text{IS} = 0.29 \pm 0.04 \text{ mm/s}$) is characteristic of a high spin iron(III) in octahedral coordination.

Upon irradiation, the spectra (Fig.2b-d) show that the initial ferric compound is progressively transformed, for increasing doses, into two high spin Fe(II) species (doublets 22' and 33') as evidenced by their large IS values ($\text{IS}_{22'} = 1.13 \pm 0.03 \text{ mm/s}$; $\text{IS}_{33'} = 1.18 \pm 0.03 \text{ mm/s}$) and their temperature dependent quadrupole splittings ($\text{QS}_{22'} = 2.35 \pm 0.03 \text{ mm/s}$, $\text{QS}_{33'} = 1.00 \pm 0.02 \text{ mm/s}$ at 298K; $\text{QS}_{22'} = 2.68 \pm 0.03 \text{ mm/s}$, $\text{QS}_{33'} = 2.03 \pm 0.03 \text{ mm/s}$ at 80K). Since these parameters (as well as their temperature dependence) are identical to those of the model compounds $\text{Cs}_2\text{Fe}(\text{ox})_2 \cdot 2\text{H}_2\text{O}$ and $\text{Cs}_2\text{Fe}(\text{ox})_2$ (Fig.2g and 2h), it can be concluded that these two compounds are present in the radiolytic product. Furthermore, as the relative areas of these two ferrous components appear to be equal whatever the dose be, the mechanism of the radiolytic decomposition may be expressed by:



This equation indicates that half of the total ferrous ions produced by radiolysis gives rise to the tetrahydrated salt whereas the other fraction remains in the anhydrous state. The above mechanism is in contradiction with a previously reported result/6/ in which the only radiolytic product of $\text{Cs}_3\text{Fe}(\text{ox})_3 \cdot 2\text{H}_2\text{O}$ was believed to be $\text{Feox} \cdot 2\text{H}_2\text{O}$, a compound characterized by Mössbauer parameters ($\text{QS} = 1.74 \pm 0.01 \text{ mm/s}$; $\text{IS} = 1.19 \pm 0.02 \text{ mm/s}$)/11/ quite different from those of the two ferrous components found in this study. Nevertheless, our mechanism is also well supported by the comparison of the TGA of the irradiated sample and of the model compounds: when heated

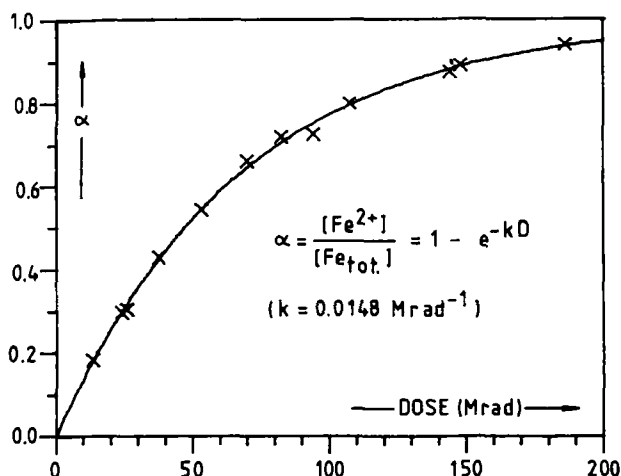


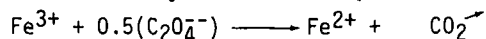
Fig.3. Iron(II) radiolytic yield as a function of the dose.

The chemical analysis data (Fig.3) show that the radiolysis proceeds as a first-order process, since the radiolytic yield α is related to the absorbed dose (H_2O) by the classical exponential law:

$$\alpha = (\text{Fe}^{2+})/(\text{Fe}^{2+} + \text{Fe}^{3+}) = 1 - e^{-k.D}$$

The corresponding least-squares calculated function is represented by the full line in Fig.3. and is characterized by an apparent rate constant k equal to 0.0148 Mrad^{-1} (if the dose is expressed in $\text{Mrad} = 10^6 \text{ rad}$). This evolution is due to the original compound depletion and to the radiolytic stability of the ferrous products.

A similar behaviour has also been found for the disappearance of the oxalate groups, which shows a saturation value $\alpha_{\text{ox}} = -(\text{ox})/\text{Fe}^{2+} = 0.5$, in agreement with the stoichiometry of the radiolytic reduction process:



At last, as the decomposition yields obtained from Mössbauer measurements (relative areas) and potentiometric titrations are similar (within $\pm 5\%$), it is likely that the recoilless processes of the various iron species are not very different in the irradiated samples, but this assumption must be taken with some caution because some annealing might have occurred during the storage of the samples in the dry nitrogen atmosphere.

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